

Alexander C. Flinchum

12101 North Lamar Blvd Apt 1021, Austin, TX 78753

alexanderflinchum@gmail.com | (317) 650-5565 | www.linkedin.com/in/alexflinchum | www.alexflinchum.dev

FULL STACK ENGINEER

Results driven Full Stack Engineer with 6+ years of experience building scalable backend services, APIs, and web applications. Core strengths include full-stack development using Java, React, TypeScript, GraphQL, C#, and PostgreSQL. Detail oriented problem solver committed to delivering reliable, high-performance software solutions from conception to production.

SKILLS

Languages: Java, Python, C#, Go, TypeScript, JavaScript, HTML5, CSS3, React.js, GraphQL, REST APIs

Frameworks: Spring Boot, .NET, Angular | **Databases & Tools:** PostgreSQL, SQL, MongoDB, Azure

Core Competencies: Object-Oriented Programming (OOP), Test Automation, Agile Methodologies, Git, Jira, CI/CD

EXPERIENCE

GENERAL MOTORS

Austin, TX

Software Engineer

Feb 2024 – May 2026

- Managed automated testing integration within GM's DaaS Java/Spring Boot platform, increasing automated regression coverage from 35% to 90% and reducing manual QA validation effort by 40% across release cycles
- Built and maintained backend services and IMOS API integrations in Java, .NET, React, and GraphQL, leveraging Copilot to support 500+ plant users across 100+ facilities managing equipment metadata and node relationships
- Spearheaded the adoption of reusable GraphQL query and service integration patterns adopted across 3+ manufacturing business units, improving backend data retrieval consistency and accelerating API onboarding for new teams by 25%
- Executed post-deployment validation scripts for 10+ services during a RedHat OpenShift migration, directly contributing to maintaining 99%+ production uptime and stability

Software Development Engineer in Test

Aug 2020 – Feb 2024

- Led the development and architecture of reusable Java and C# test automation frameworks from scratch, reducing manual regression testing by 60% and cutting defect escape rates by 25% across 8+ bi-weekly release cycles annually
- Directed end-to-end release readiness coordination across a 5+ cross-functional team, resolving 15+ production and integration defects per sprint and maintaining enterprise application stability
- Drove the expansion of automated test coverage from 30% to 85% across enterprise backend integrations and application workflows, accelerating QA cycle time by 2 days per sprint and enabling more reliable bi-weekly deployments

Software Developer

Jul 2019 – Aug 2020

- Headed the maintenance of InDepth Lite, an Angular-based file-processing application supporting 150 users across GM's analytics platform, collaborating with frontend, backend, and analytics teams to sustain 99%+ reliability
- Supervised the modernization of 100% of legacy Angular components during a 3-month framework migration, resolving deprecated dependencies and regressions with zero disruption to active production workflows
- Guided the identification and resolution of 15+ frontend UI defects, reducing user-reported support tickets by 30% and improving application stability across active development cycles

EDUCATION

BOSTON UNIVERSITY

Boston, MA

Master's of Science in Software Development

May 2025

- Coursework:** Data Structures and Algorithms, Web Application Development, Software Design and Patterns, Advanced Programming Techniques, Database Design and Implementation for Business, Software Engineering

INDIANA UNIVERSITY

Bloomington, IN

Bachelor's of Science in Informatics (Distinction)

May 2019

Minor in Computer Science | Cognate in Game Design | GPA: 3.8/4.0

- Honors:** National Society of Collegiate Scholars, Hutton Honors College Inductee
- Awards:** Dean's List Honors, 3-time Recipient of Indiana University Founders Scholar

PROJECTS

FitQuest Backend

- Designed and developed a scalable Go backend using GraphQL and MongoDB, leveraging OpenCode, with decoupled packages and the repository pattern, enabling clean unit testing and rapid integration of new exercise models in under 2 hours
- Implemented secure JWT authentication and role-based access control across 100% of API endpoints to safeguard user data